Pool Game Player Guide

How to run the game in PC mode and VR mode

Overview

This project has two ways to play: a PC mouse-and-keyboard mode and a VR mode. The game starts in PC mode by default, so a new player can open it and begin playing right away. Press V at any time to switch into VR mode.

The goal is simple: aim the cue stick, set your shot power, and hit the cue ball. The on-screen HUD shows the current turn, the remaining balls for each player, fouls, and the current power level.

How to start

1. Open the Unity project and press the Run / Play button.

   OR

   1.Go to the **Build folder** of the project. Open the .exe file (*Pool Game(VR).exe*).

2. The game will launch in the PC mouse-and-keyboard version.
3. Use the mouse to aim the cue stick.
4. Use the on-screen controls or the keyboard shortcuts listed below.
5. Press V whenever you want to switch into VR mode.

PC mouse and keyboard mode

This is the default mode shown in the first screenshot. It is designed for quick testing and for players who want to use a mouse and keyboard instead of a headset.

PC mode controls

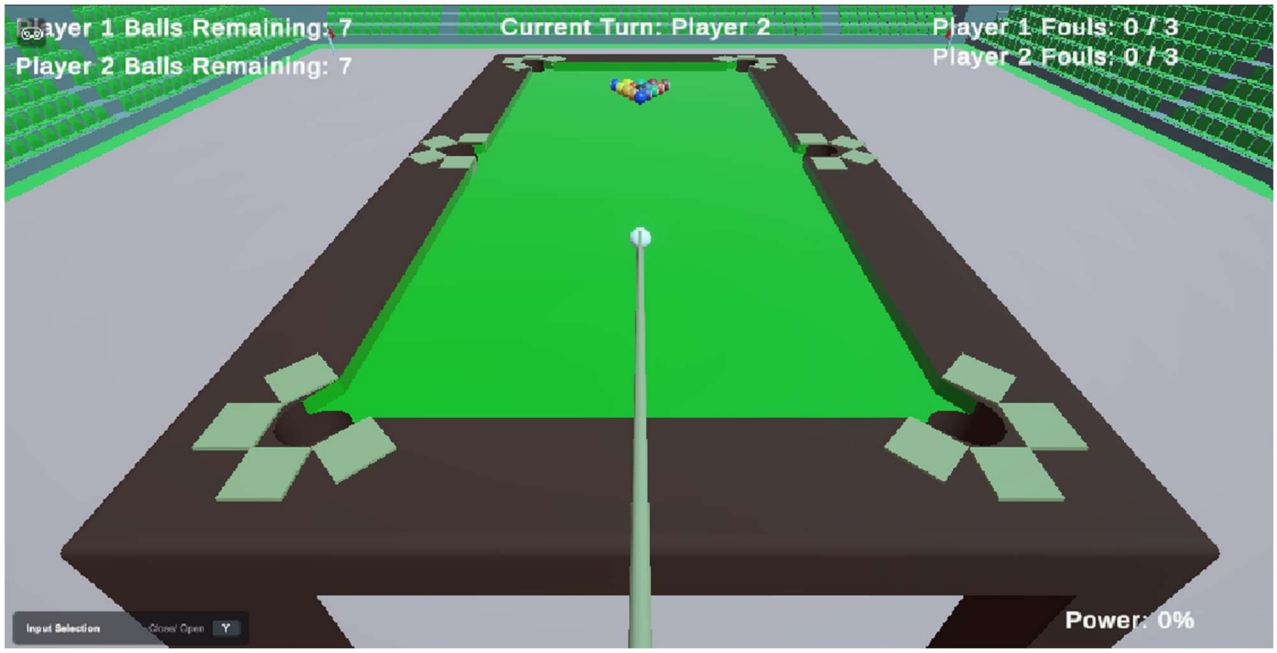
Key / Action	What it does
Mouse movement	Move the mouse to change the direction of the cue stick and aim at the cue ball.
Left mouse drag	Drag to control shot power from 0% to 100%. The power indicator is shown at the bottom right.
Space	Reset the camera and place the cue stick back in front of the cue ball.
Top-left controller button	Switch to the overhead camera view.
V	Switch from PC mode to VR mode.
Y	Open or close the Input Selection panel.

In PC mode, the player can move the mouse to line up the shot, then adjust the power before taking the strike. This mode is useful for learning the game and for testing the logic without using a headset.

On-screen HUD(Heads-Up Display) in PC mode

The first screenshot shows the main HUD. These labels are displayed during play:

- Player 1 Balls Remaining
- Player 2 Balls Remaining
- Current Turn
- Player 1 Fouls
- Player 2 Fouls
- Power: 0% to 100%



PC mode view with cue aiming and power indicator.

VR mode

Press V to switch into VR mode. In VR, the player gets the full headset experience, while the keyboard shortcuts still remain available for movement and testing.

VR movement and shot controls

Key	Action
Arrow keys	Rotate the cue stick angle and adjust the camera orientation.
W / A / S / D	Move the cue stick forward, left, back, and right.
Q / E	Move the cue stick up and down.
G	Hit the cue ball.
Space	Place the cue stick in front of the cue ball again, or reset the camera.

The VR control scheme is mapped so the game can still be played even when the headset is not available. That makes it easier for a new player to understand the game flow before using the headset.

Controller Actions panel

The second screenshot shows the in-game control help. This panel lists the main controller actions and the matching hotkeys.

Shown on screen	Meaning
Trigger -> T	Trigger action.
Grip -> G	Grip action.
Primary -> 1	Primary controller action.
Secondary -> 2	Secondary controller action.
Thumbstick -> I / J / K / L	Thumbstick movement mapped to the I, J, K, and L keys.
Press / Release -> Space	Press or release action.
Cycle Actions -> `	Cycle through available actions.
Reposition into view -> R	Bring the controller or view back into a comfortable position.
Close / Open -> X	Open or close the Controller Actions panel.

Input Selection panel

The Input Selection panel is shown in the second screenshot as well. It helps the player switch between body, hand, and head-based input.

Shown on screen	Meaning
Close / Open -> Y	Open or close the Input Selection panel.
Full Body -> Tab	Switch to full-body input.
Left bracket [	Select left-hand access.
U	Switch to both-hands access.
] Right bracket	Select right-hand access.
H	Select head input.
Rotate -> Arrow keys	Rotate the selected input.
Translate -> Q / W / E and A / S / D	Move the selected input in space.

Input Selection and Controller Actions reference panel.

Quick tips for a first-time player

- Start in PC mode and get comfortable with aiming before switching to VR.
- Watch the power indicator so the shot is not too weak or too strong.
- Use the overhead camera if you need a better view of the table.
- Use the help panels whenever you forget a shortcut.
- If the headset is not available, the keyboard mapping still lets the game run cleanly for testing and demonstration.